Nick Schutts – Engineering Physics

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PROFILE

Second-year Engineering Physics student with hands-on experience in electromechanical design and integration through Formula SAE. Proficient in CAD, with a demonstrated ability to learn new tools quickly and adapt to difficult and novel problems. A steady, outgoing communicator with experience working in fast-paced settings and collaborating with diverse teams.

SKILLS

Mechanical: SolidWorks/PDM, 3D Printing, Laser/Waterjet Cutting, Lathe Turning, GD&T, Altair FEA
Software: Altium Designer, MATLAB, C/C++, Java, Git, LTSpice, Python, MS Excel, VS Code
Electronic: Arduino, Soldering, Oscilloscopes, Breadboard, STM32, PCB Design, Power Management

TECHNICAL PROJECTS AND EXPERIENCE

Electrical Subteam Member **Sep 2025 - Present**
Formula UBC Racing · Vancouver, BC

Formula SAE Dashboard Project Portfolio

- Engineered innovative 3D-printed dashboard enclosure for UBC's 2026 Formula SAE car using SolidWorks, ensuring electrical compatibility as well as proper integration with other vehicle subsystems.
- Validated design through various prototyping methods and project iterations, while also adapting to repeated changes in design parameters.

Vehicle System Validation Board (VSVB) · *In Progress* Portfolio

- Designing PCB validation unit for system level benchtop testing of FSAE wiring harness.
- Interfacing directly with Haltech Elite 1500 ECU, PDM, and dashboard to simulate analog, digital, and reluctor sensor inputs for drive-by-wire implementation.
- Developing firmware using C++ wrapped around CMSIS_V2 FreeRTOS to manage real-time task scheduling and signal simulation.

Autonomous Claw Project Portfolio

- Independently produced electromechanical design for in-class claw retrieval competition, including electrical circuit schematics for Arduino integration.
- Implemented guidance program in C++, used to pilot the claw autonomously during competition.

WORK EXPERIENCE

Bartender **Apr 2026 - Sep 2026**
Northview Golf and Country Club · Surrey, BC

- Worked individually in a fast-paced, high-stress environment, successfully prioritizing multiple concurrent tasks and strict time constraints.
- Maintained high attention to detail, ensuring consistently high drink and service standards.

- Optimized organization and inventory, using product knowledge to facilitate communication between guests, servers, kitchen staff, and management.

Pharmacy Assistant **Feb 2023 - Aug 2025**
Northmount Compounding Pharmacy · North Vancouver, BC

- Provided professional customer service to patients, both over the phone and in store.
- Collaborated with pharmacists in a fast-paced environment, helping to accurately fill up to 225 prescriptions per day.
- Managed inventory, drug ordering, prescription filing, expired medication returns, and courier deliveries.

EDUCATION

Bachelor of Applied Science, Engineering Physics **Sep 2025 - Present**
University of British Columbia · Vancouver, BC **GPA: 93.2%**

- Trek Excellence Scholarship (2025/26)
- Dean's Scholar (2025/26)

BC Dogwood Diploma **Sep 2021 - Jun 2025**
Vancouver Waldorf School · North Vancouver, BC